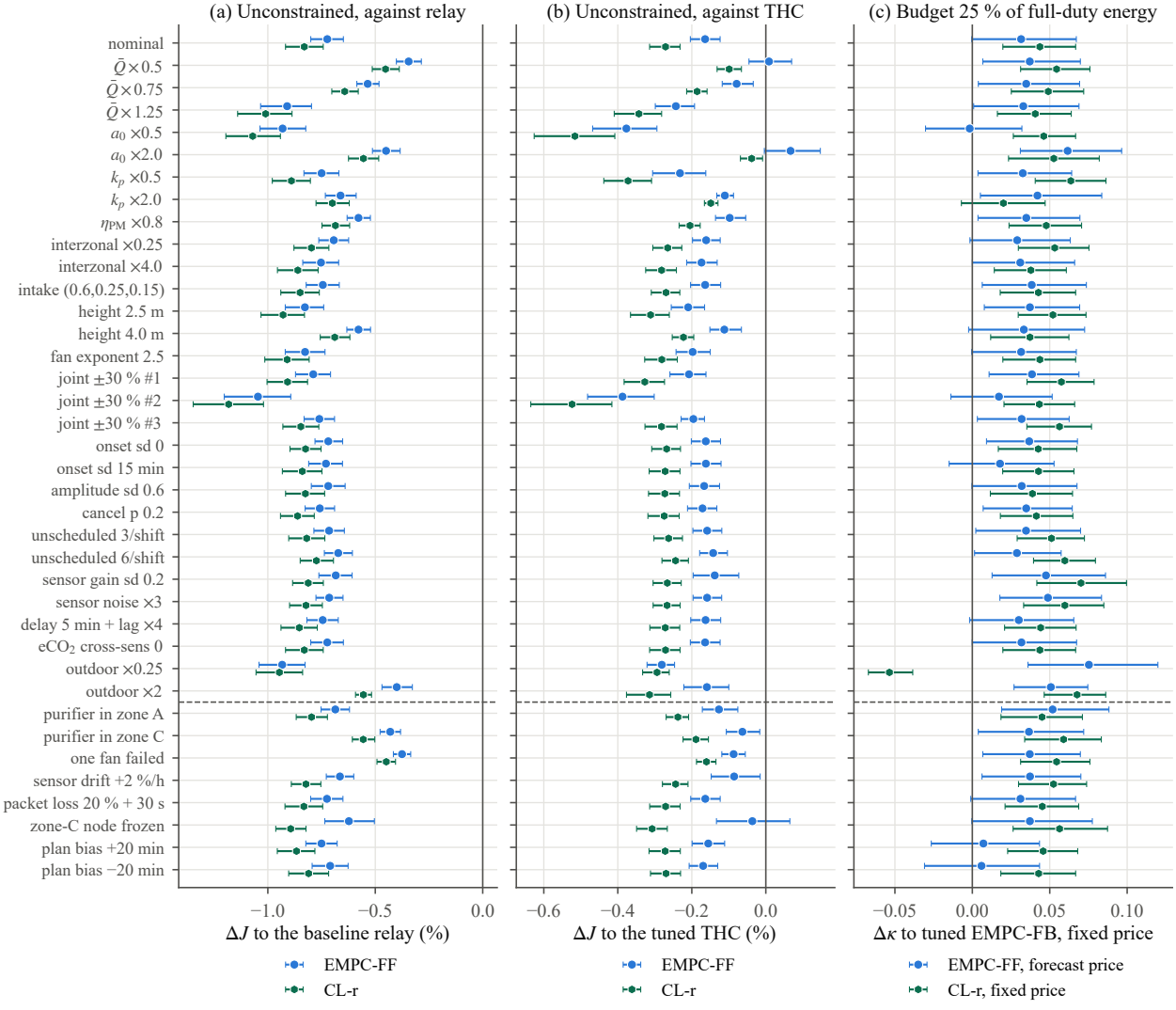

(a) Unconstrained, against relay
(b) Unconstrained, against THC
(c) Budget 25 % of full-duty energy
nominal
$\bar{Q}\times0.5$
$\bar{Q}\times0.75$
$\bar{Q}\times1.25$
$a_0\times0.5$
$a_0\times2.0$
$k_p\times0.5$
$k_p\times2.0$
$\eta_{PM}\times0.8$
interzonal ×0.25
interzonal ×4.0
intake (0.6,0.25,0.15)
height 2.5 m
height 4.0 m
fan exponent 2.5
joint ±30 % #1
joint ±30 % #2
joint ±30 % #3
onset sd 0
onset sd 15 min
amplitude sd 0.6
cancel p 0.2
unscheduled 3/shift
unscheduled 6/shift
sensor gain sd 0.2
sensor noise ×3
delay 5 min + lag ×4
eCO_2 cross-sens 0
outdoor ×0.25
outdoor ×2
purifier in zone A
purifier in zone C
one fan failed
sensor drift +2 %/h
packet loss 20 % + 30 s
zone-C node frozen
plan bias +20 min
plan bias −20 min
−1.0
−0.5
0.0
−0.6
−0.4
−0.2
0.0
−0.05
0.00
0.05
0.10
ΔJ to the baseline relay (%)
ΔJ to the tuned THC (%)
Δκ to tuned EMPC-FB, fixed price
EMPC-FF
CL-r
EMPC-FF
CL-r
EMPC-FF, forecast price
CL-r, fixed price